

n8n AI Agent Setup Guide

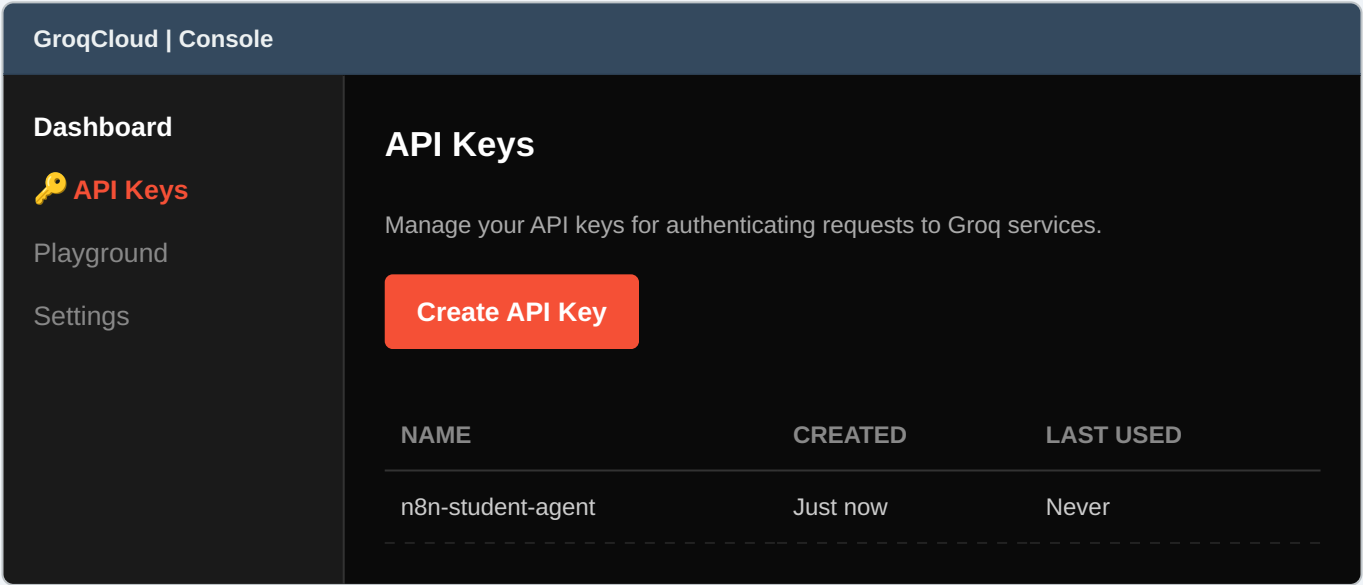
Integrating Groq and Hugging Face

This document provides comprehensive instructions and high-fidelity interface diagrams to guide you through creating free student accounts on Groq and Hugging Face, generating API keys, and configuring headless AI Agents within n8n using custom temperature settings.

Part 1: Groq Integration

1. Creating an Account & Generating the API Key

1. Navigate to **console.groq.com** and sign up using your student email or GitHub account.
2. Once logged into the Groq Cloud Console, locate the left-hand navigation sidebar.
3. Click on **API Keys**.
4. Click the prominent **Create API Key** button.
5. Assign a descriptive name (e.g., n8n-student-agent) and click Submit.
6. **Copy the generated API key immediately.** For security reasons, Groq will not display it again.



2. Configuring Groq Credentials in n8n

1. Open your n8n instance and navigate to the **Credentials** section via the left sidebar.
2. Click **+ Add Credential** in the top right.

3. Search for **Groq API** and select it.
4. Paste the API key you copied from the Groq console into the **API Key** field.
5. Rename the credential to Groq Student API Key and click **Save**.

n8n | Credentials

Groq API

Name

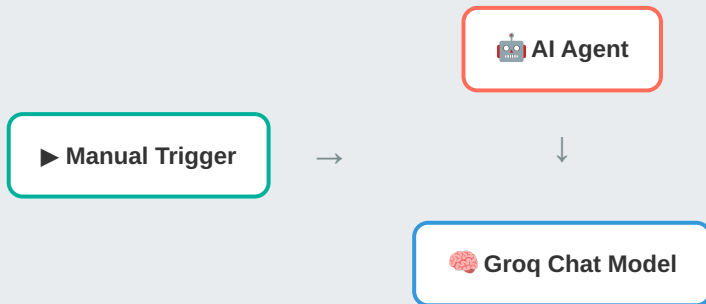
API Key

Save

3. Configuring the n8n Agent

We will construct a headless AI Agent (no tools, no memory) driven by a manual trigger.

1. In a new workflow, add a **Manual Trigger** node (On clicking 'Execute').
2. Add an **AI Agent** node and connect the trigger output to it.
3. Double-click the AI Agent. Ensure "Agent Type" is standard (e.g., Conversational). **Leave Memory and Tools disconnected.**
4. Hover over the "Model" input at the bottom of the AI Agent node and click +.
5. Select **Groq Chat Model**.
6. In the Groq Model properties, select your Groq Student API Key. Select a free model like llama3-8b-8192.
7. Under Options, add **Temperature** and set it to 0.7.
8. **To Test:** Enter a prompt in the AI Agent's Text field (e.g., *"Explain an API to a high schooler."*) and click **Test Workflow**.



Groq Chat Model

Credential for Groq API

Groq Student API Key ▼

Model

llama3-8b-8192 ▼

Options

Temperature

0.7

Part 2: Hugging Face Integration

1. Creating an Account & Generating the Access Token

1. Navigate to **huggingface.co** and click Sign Up.
2. Once registered and logged in, click your profile picture in the top right corner and select **Settings**.
3. In the left sidebar, navigate to **Access Tokens**.
4. Click **New token**. Name it n8n-inference.
5. Select **Read** or **Fine-grained** permissions (ensure Inference API access is checked).
6. Click Generate and copy the token (it will begin with hf_).

 Hugging Face

[Profile](#)
[Account](#)
[Access Tokens](#)
[Billing](#)

Access Tokens

Tokens for authenticating API requests and accessing private models.

[Create new token](#)

NAME	ROLE	TOKEN
n8n-inference	Read	hf_*****

2. Configuring n8n Agent for Hugging Face

1. In n8n, create a new credential for **Hugging Face API** and paste your hf_ token. Save it as HF Student Token.
2. In your workflow canvas, add an **AI Agent** node connected to a Manual Trigger. (Do not connect tools or memory).
3. Hover over the "Model" input of the AI Agent node, click +, and select **Hugging Face Chat Model**.
4. Select your HF Student Token credential.
5. Enter an open-source inference model ID, such as meta-llama/Meta-Llama-3-8B-Instruct.
6. Under Options, add **Temperature** and set it to 0.5.

7. **To Test:** Double-click the AI Agent, enter a test prompt (e.g., *"Write a haiku about n8n."*), and execute the node. Ensure you check the output to verify successful inference!

Note: Hugging Face's free serverless Inference API can experience "cold starts". If your first manual execution times out or fails, wait 30 seconds and test the workflow again. The model takes a moment to load into memory.